

Context ameliorates but does not eliminate garden-pathing: Novel insights from latent-process modeling

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Abstract

A theoretically important claim in psycholinguistics is that in English, linguistic context largely eliminates garden-path effects in temporary complement clause/relative clause ambiguities (e.g., Altmann, Garnham, and Dennis, 1992, Avoiding the garden path: Eye movements in context. *Journal of Memory and Language*, 31 (5), 685–712). However, this strong claim, based on comparing mean differences between groups of conditions in a factorial design, reduces the conclusion to a overly simplistic binary one: either context affects garden-pathing or it doesn't. Mean differences are an aggregate of different, non-deterministic parsing steps that could be very different from trial to trial. As a result, there may not be a simple yes-no answer to whether context eliminates garden-pathing. Sentence comprehension has been argued to involve a cascade of probabilistically occurring latent processes (e.g., Paape and Vasishth, 2022); seen in this way, context-sensitivity may only be occurring probabilistically and in a graded manner. We present a hierarchical multinomial processing tree model that spells out several probabilistic latent processes. The model reveals new insights into the impact of context on garden-pathing, going beyond the simple question of whether context reduces processing difficulty *on average*. Modeling using new data from a bidirectional self-paced reading study with 319 participants shows that the context affects both first-pass attachment probability and reanalysis cost. Because a simpler account of context effects than the MPT model could be a model based on surprisal derived from large language models, we compare the MPT model's predictive fit to a model using surprisal computed from large language models. Results show that the MPT model, which assumes a mixture distribution of reading times, outperforms surprisal-based models at predicting garden-path effects.

Keywords: *garden-path effect, syntactic reanalysis, multinomial processing tree, latent processes, mixture modeling*

Introduction

Does discourse context modulate first-pass parsing choices in cases of syntactic ambiguity? Consider the two discourses in the following example from Altmann et al. (1992). In the temporarily ambiguous target sentence, the substring *that he'd risked his life for* can initially attach either as a sentential complement of the verb *told* or as a relative clause modifying *the woman*:

To avoid choosing the incorrect analysis, it would make sense for the human parser to consider the discourse context when deciding where to attach the material following *that*: a relative clause would help distinguish between the two women in the case of non-unique referents but not in the case of the unique referent. By contrast, in the case of a unique referent, the relative clause attachment to *the woman* would lead to a case of presupposition violation, given that there is no unique woman to refer to (Altmann and Steedman, 1988). We will call this the *context-sensitive attachment* hypothesis.

Despite the apparent utility of using context information to inform first-pass parsing choices, a case can also be made that the parser should always avoid the relative clause analysis. Compared to the complement clause analysis, the relative clause analysis requires the stipulation of additional syntactic structure to make a complex noun phrase ($[[\text{the woman}]_{NP} [\text{that} \dots]]_{NP}$).¹ Furthermore,

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¹Another added complexity is that the relative clause structure is typically analyzed as requiring syntactic movement, and thus involving a gap after the embedded verb (e.g., Staub et al., 2018).

CONTEXT	
Unique referents	Non-unique referents
An off-duty fireman was talking to a man and a woman.	An off-duty fireman was talking to two women.
He was telling them how serious the situation had been when their house had caught fire.	
The fireman had risked his life to rescue the woman while the man had waited outside.	The fireman had risked his life to rescue one of the women while the other had waited outside.
TARGET SENTENCE	
Complement clause	
He told the woman <u>that he'd risked his life for many people in similar fires.</u>	
Relative clause	
He told <u>the woman that he'd risked his life for</u> to install a smoke detector.	

Table 1: The experimental design of Altmann et al. (1992) (ambiguous conditions only).

relative clauses are optional, while introducing the sentential object argument of the verb *told* is mandatory. We will call this the *context-insensitive attachment* hypothesis.

From a theoretical point of view, it is an open question which type of processing pressure dominates in shaping the parser's behavior. The question is especially relevant given the risk that the chosen structure later turns out to be incorrect, which would force the parser to revise its analysis. In such cases, a garden-path effect should be visible in the reading record. The empirical question is whether garden-pathing as indexed by reading disruption — that is, longer reading times at the disambiguating region and/or regressions followed by rereading — is modulated by the linguistic context or not. The context-sensitive attachment hypothesis predicts that any mismatch between the linguistic context and the type of disambiguation should result in reading disruption, as the parser's context-informed first-pass analysis will turn out to be incorrect more often if the continuation doesn't match the context. By contrast, the context-insensitive attachment hypothesis predicts that the relative clause disambiguation should cause more reading disruption regardless of context, as the parser prefers the complement clause analysis for reasons of parsimony.

An influential proposal that would appear to align with context-sensitive attachment is surprisal theory, which argues that a word's processing difficulty is a function of its predictability in context, more precisely of its negative log probability (Hale, 2001; Levy, 2008). Surprisal theory in its formulation by Levy (2008) assumes that ambiguity resolution is parallel and nondeterministic: possible structures are ranked according to their plausibility given previous input, and processing difficulty increases when the ranking changes in response surprising continuations. Garden paths are not inherently special under this view, given that they can be seen as instances of such a costly reranking taking place at the point of disambiguation. Indeed, surprisal has been argued to be the only variable needed to predict any given word's processing difficulty (Smith and Levy, 2013). In recent years, it has become the norm to compute surprisal based on probabilities extracted from large language models (LLMs) such as GPT, and to evaluate how well human reading times and/or other processing measures such as ERP amplitudes align with LLM surprisal (e.g., Wilcox et al., 2020; Shain et al., 2024; Krieger et al., 2024). Recent work has indeed shown that LLMs can mimic human behavior in many psycholinguistic experimental designs (Cai et al., 2024), though they do not fully replicate human sensitivity to linguistic context (Britton et al., 2024; Hanna et al., 2023; Michaelov et al., 2023).

LLMs learn to predict words based on statistical patterns present in their linguistic input. Given a large enough context window, they should thus, in principle, be capable of learning any pattern in which one piece of linguistic information predicts another piece of linguistic information further downstream. With regard to the current work, if there is a correlation in the language model's training data between the linguistic context containing unique versus non-unique referents (*a man and a woman* versus *two women*) and the subsequent use of a relative clause or complement clause (*the woman that ...*), the models should presumably be able to learn this relationship, and their surprisal

values for the different disambiguations should change accordingly, in line with context-sensitive attachment.

In human participants, existing empirical results regarding context effects on syntactic disambiguation are decidedly mixed.² Crucially, when judging the literature, it is important to keep in mind that the context-sensitive attachment hypothesis predicts a statistical interaction between context and disambiguation, while the context-insensitive attachment hypothesis in its strongest form predicts the *absence* of such an interaction. Depending on the statistical power of the experiment, the mere absence of a significant interaction does not necessarily mean that there truly is none (e.g., Altman and Bland, 1995). Furthermore, there is the question of which reading measures should even be considered to index garden-pathing during first-pass processing, as context may also influence “early” second-pass processing (Rayner and Sereno, 1994; Clifton Jr and Ferreira, 1989). Studies may also use different stimulus materials, which can affect the effectiveness of the context manipulation (Altmann, 1994; Staub et al., 2018).

Perhaps most importantly for our purposes in this paper, there is the possibility of a mixed pattern of results that is compatible with *some* graded amount of context-sensitivity: on average, relative-clause (RC) disambiguation may be associated with more disruption during first-pass processing than complement-clause (CC) disambiguation — this would be a main effect of disambiguation type — but at the same time the disruption may be modulated by contextual match to some degree, either because first-pass attachment choices are nondeterministic or because reanalysis is easier in an RC-supporting context. This is the pattern that was actually observed in the eye-tracking studies of Altmann et al. (1992). While first-pass reading times showed no evidence of disruption for RC disambiguation when the context introduced non-unique referents (that is, *two women* in Table 1), it *did* still lead to more first-pass regressions from the disambiguating region compared to complement-clause disambiguation, but the percentage of regressions was reduced in matching contexts. Altmann et al. concluded that “but for a small minority of occasions, felicitous referential context eliminates the garden paths that are otherwise found with the ambiguous relatives in the [mismatching] context” (p. 704).

Altmann et al.’s experiments have, to our knowledge, never been directly replicated. The only study that used a similar design that crossed contextual support and disambiguation was a grammaticality rating study by Crain and Steedman (1985, Expt. 2), who found a crossover interaction: when the disambiguation did not match the context, both relative-clause and complement-clause continuations were rejected about 50% of the time (compared to 22%/12% in the matching conditions). Other studies either compared RC and CC disambiguation without a context manipulation (Staub et al., 2018; Vernice et al., 2016; Demestre and García-Albea, 2004; Mitchell et al., 1992), or investigated context effects for a different type of syntactic ambiguity (VP/NP attachment: Altmann and Steedman, 1988; main clause/reduced relative clause: Spivey-Knowlton et al., 1993; Spivey and Tanenhaus, 1998; high/low attachment of relative clauses: Dempsey et al., 2023; subject/object relatives: Altmann et al., 1994; Fedorenko et al., 2012).

The most recent and well-powered eye-tracking study ($n = 160$) by Dempsey et al. (2023) on high versus low attachment of relative clauses (*The daughter of the actor who devoted herself . . .*) in matching, neutral and mismatching contexts showed contextual mismatch effects in regression probabilities, go-past times, and rereading times, while the results for first-fixation durations and first-pass reading times were inconclusive. Dempsey et al. concluded that their findings were more compatible with context-insensitive first-pass attachment, followed by a monitoring process that checks for compatibility of the assigned structure with the context (a “weakly interactive” system in the terminology of Grodner et al., 2005). However, the inconclusive results for the “early” eye-tracking measures leave open the possibility that the system is, in fact, “strongly interactive.” Below, we aim to shed more light on this question by conducting a replication of the Altmann et al. (1992) study, and fitting a computational model to the data whose purpose is to disentangle the latent processes that contribute to garden-path effects.

²We limit our discussion to temporarily ambiguous structures. For a recent discussion and experimental results regarding possible contextual influences on the processing of unambiguous structures, see Dempsey and Christianson (2022).

Current study

Computational modeling approach

The main motivation for the current study lies in the fact that the standard approach of comparing the means of different reading measures across conditions is bound to yield an overly simplistic picture of what is actually happening during garden-pathing and subsequent recovery. For instance, the question of whether linguistic context affects first-pass parsing choices, the ease of recovery from a misanalysis, or possibly both, is difficult to answer based on average reading times across conditions (Rayner and Sereno, 1994; Clifton Jr and Ferreira, 1989). In order to disentangle the different components of garden-pathing, Paape and Vasishth (2022b) have recently proposed a Bayesian computational model that assumes a cascade of probabilistically occurring latent subprocesses, and that takes into account key insights and proposals from the literature:

1. Reading data is contaminated by trials in which the participant is not paying attention (e.g., Smallwood, 2011; Schad et al., 2012). In such trials, participants will show fast reading times, but will resort to guessing in whatever end-of-trial task is used (e.g., comprehension questions, acceptability judgments).
2. Parsing choices are not deterministic but stochastic at the trial level (e.g., van Gompel et al., 2000; Jurafsky, 1996). In trials where the correct analysis is adopted on the first pass, there will be no garden-pathing.
3. The processing cost of garden-pathing itself is distinct from the cost of reanalysis. In some trials, readers are garden-pathed but do not carry out reanalysis, “triaging” the parse instead and rejecting the sentence as ungrammatical (Meng and Bader, 2000; Fodor and Inoue, 2000).
4. Reanalysis can be carried out overtly, which is signaled by rereading of earlier material, but also covertly, which is signaled by an *in situ* reading slowdown at the disambiguating region (Lewis, 1998).
5. Reanalysis can fail, which results in the sentence being rejected as ungrammatical (Warner and Glass, 1987).

The basic premise behind the Paape and Vasishth (2022b) model is that reading times at the disambiguating region are a mixture of multiple components that reflect different combinations of latent processes for each experimental trial. The reading measures at the disambiguating and spillover regions of the sentence are also assumed to be intrinsically connected to participants’ responses to end-of-trial tasks. Mathematically, instead of having a single distribution with a single mean, reading times in each condition are modeled as coming from a mixture distribution with multiple means (e.g., Vasishth et al., 2017; Nicenboim and Vasishth, 2018; Lissón et al., 2021). The mixing proportions of this distribution correspond to the probabilities of following different processing paths in a multinomial processing tree (MPT), a common class of model in mathematical psychology (see Erdfelder et al., 2009; Batchelder and Riefer, 1999 for reviews). The chosen path through the tree also determines whether a first-pass regression is observed in a given trial, and determines the response to the end-of-trial task. For a given tree, the parameters can be inferred from the data via the laws of probability: given a certain proportion of trials in each “bin,” the product rule and the law of total probability can be applied to recover the process probabilities as well as any effects of the experimental manipulations on these probabilities.

In theory, all process probabilities and costs in the model can be affected by whatever manipulations are used in the experiment. This amounts to fitting a regression model with a slope for the manipulation in question for the parameter of interest, such as the probability of choosing the correct parse on the first pass or the cost of reanalysis. In the present case, we are interested in effects of the context manipulation (CC-supporting versus RC-supporting) and the disambiguation manipulation (CC disambiguation versus RC disambiguation versus unambiguous, see below). Process probabilities and costs can also differ between participants and between individual sentences, which are modeled as random effects using a partial-pooling approach (Singmann et al., 2024).

Paape and Vasishth (2022b) made some simplifying assumptions in their model: they assumed that the cost of covert reanalysis is constant across constructions and that this cost is always paid at the disambiguating region; and they did not take into account non-decision time (Rouder, 2005; Wagenmakers et al., 2007). In the current work, we extend the MPT model by incorporating three additional proposals from the psycholinguistics and decision-making literature:

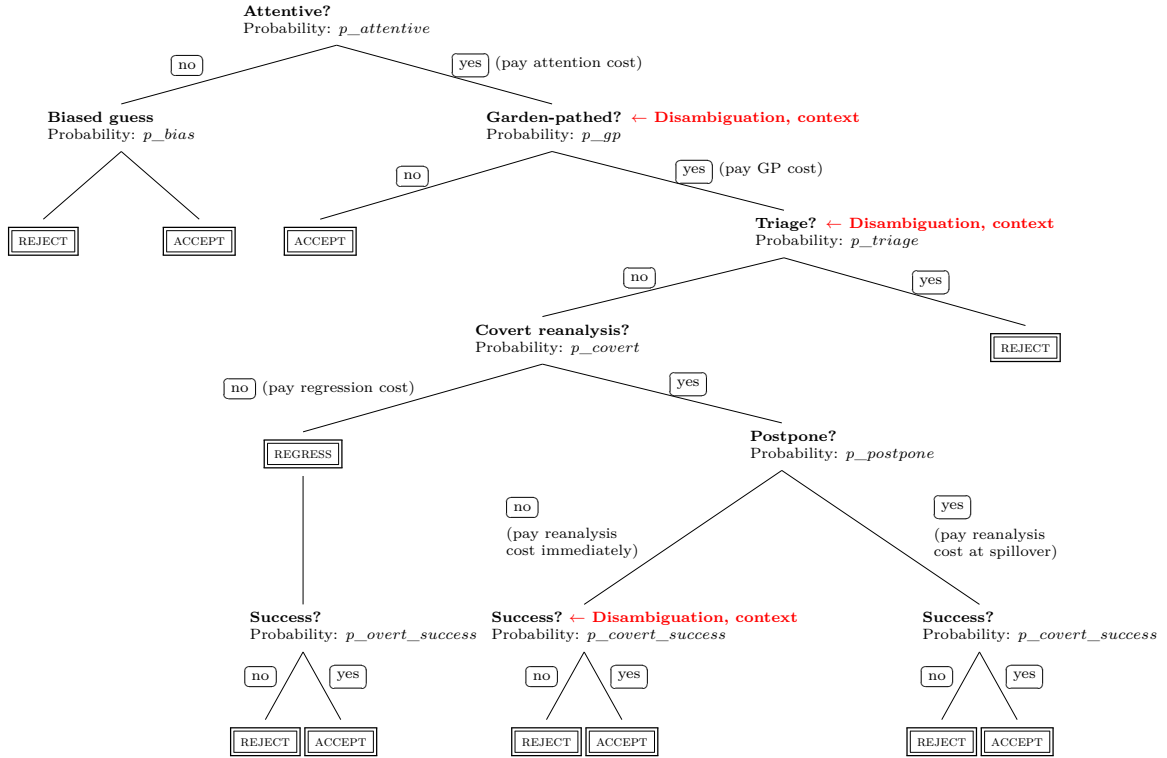


Figure 1: The extended multinomial processing tree model of garden-pathing and reanalysis.

6. Covert reanalysis can be postponed, in which case the associated reading slowdown will be registered at the spillover region instead of the disambiguating region (Keshev and Meltzer-Asscher, 2020; Tily et al., 2010).
7. The cost of covert reanalysis may differ based on the direction of the structural change (RC-to-CC vs CC-to-RC; e.g., Fodor and Inoue, 1998).
8. Response time distributions are shifted to the right: each data point contains a certain amount of “non-decision” time that reflects motor and encoding processes and that is independent from the stimulus response (e.g., Rouder, 2005; Wagenmakers et al., 2007).

Figure 1 shows the extended MPT model for an experiment with end-of-trial grammaticality judgments.

In the next section, we first report the results of an experiment that seeks to replicate the findings of the original Altmann et al. (1992) study. In order to facilitate comparisons of our results with those of the original study, we first report standard statistical analyses of the theoretically interesting reading measures. Then we turn to the main focus of the present work, the computational modeling results.

Experimental data

Stimuli

The 36 stimulus sentences were adapted from Altmann et al. (1992). Changes mainly involved removing cultural references specific to the United Kingdom, such as references to the Royal Family, as well as changing UK to US spelling. Furthermore, Altmann et al.’s stimuli disproportionately used the verb *told*, which has a strong bias towards taking sentential complements (Staub et al., 2018). We thus replaced *told* with different verbs in all except seven items. Where necessary, sentences were also adapted such that there was at least one spillover region between the disambiguating region and the sentence-final region. An example item with presentation regions separated by the “◊” symbol and the disambiguating region printed in bold is shown in (1). In addition to the RC- and

CC-disambiguation conditions, there was also a control condition with a verb that doesn't allow CC continuations, such as *asked* in (1c).

- (1) (a) Ambiguous, CC disambiguation
He explained ◊ to the woman ◊ that ◊ he'd risked his life ◊ for ◊ **many people** ◊ before ◊ in similar fires.
- (b) Ambiguous, RC disambiguation
He explained ◊ to the woman ◊ that ◊ he'd risked his life ◊ for ◊ **that she needed** ◊ to install ◊ a smoke detector.
- (c) Unambiguous, RC disambiguation
He asked ◊ the woman ◊ that ◊ he'd risked his life ◊ for ◊ **to install** ◊ a smoke detector ◊ and a sprinkler system.

Following Altmann et al. (1992), the disambiguation manipulation was crossed with the context manipulation from Table 1 in a 3×2 design ({amb-CC, amb-RC, unamb-RC} × {unique ref., non-unique ref.}). There were also 36 filler sentences that, like the critical items, were presented as discourses. Some filler discourses were well-formed, while others featured contextual mismatches between the use of *only some* and *all* (e.g., *Only some of the stocks went up. In fact, they all did ...*; Bergen and Grodner, 2012), or contained other kinds of contextual mismatches (e.g., *Rover barks at all passing animals. He's the most alert kitten ...*; Zargar et al., 2020).

0.0.1 Participants

319 self-reported native speakers of English were recruited on Prolific. They were paid £3.80 for their participation. Pre-screening was implemented so that the study only recruited participants who reported English as their first and primary language, were currently living in the US or UK, had normal or corrected-to-normal vision, and had no diagnosed or suspected dyslexia.

Procedure

Even though the original experiment by Altmann et al. (1992) used eye tracking while reading, we instead opted to conduct our experiment using masked bidirectional self-paced reading (BSPR). The BSPR methodology yields measures that closely resemble eye-tracking data, and previous work (Paape and Vasishth, 2022c,a; Witzel and Witzel, 2023) has demonstrated that BSPR can be used successfully to investigate garden-path phenomena. As in other methods, such as classical self-paced reading (Just et al., 1982) and the Maze task (Boyce et al., 2020), there is a trade-off between the naturalness of the task and the need for achieving reasonable statistical power. Although BSPR is less naturalistic than eye tracking due to the impossibility of skipping and due to regressions being consciously triggered (see below), it is significantly less resource-intensive, and offers the huge advantage of being easy to administer remotely, giving us access to a larger pool of English native speakers.

In BSPR, participants move a sliding presentation window through the sentence. Unlike in classical self-paced reading, participants can move the presentation window backward instead of forward at any point. This presentation mode allows for the computation of reading measures familiar from eye-tracking research, such as first-pass reading time, percentage of first-pass regressions, and total reading time. A sample trial is shown in (2).

- | | | | | | |
|-----|------|----------------------|------------|-------|---------------------------------------|
| (2) | that | _____ | --- | _____ | Key press: → |
| | --- | he'd risked his life | --- | _____ | Key press: → |
| | --- | _____ | for | _____ | Key press: → |
| | --- | _____ | to install | _____ | Key press: ← (first-pass regression) |
| | --- | _____ | for | _____ | Key press: ← (second-pass regression) |
| | --- | he'd risked his life | --- | _____ | Key press: ← (second-pass regression) |

In addition to moving forward and backward through the sentence with the arrow keys, participants can use the escape key to return to the beginning of the sentence, or use the control key to jump directly to the end-of-sentence acceptability judgment.

Participants were informed about the nature and general purpose of the study and gave their informed consent. After each trial, participants judged whether the final sentence of the discourse

was acceptable in the context. There were two practice trials. Participants' mean completion time for the experiment was 26 minutes.

Results and discussion

Figure 2 shows first-pass reading times by context, condition, and region. Figure 3 shows the corresponding percentages of first-pass regressions. Figure 4 shows the percentages of positive acceptability judgments by condition.

We carried out a standard statistical analysis using Bayesian linear mixed-effects models to ensure comparability with previous studies. The detailed results, which include Bayes factors calculations (Kass and Raftery, 1995) for the effects discussed below, are presented in the appendix. We also conducted a scanpath analysis (von der Malsburg and Vasishth, 2011; von der Malsburg, 2018) to identify selective rereading strategies during overt reanalysis, which however failed to yield any informative results.

Summarizing the results, the general picture is that in both first-pass reading times and first-pass regressions, RC disambiguation leads to more processing difficulty at the critical disambiguating region compared to CC disambiguation and to the unambiguous RC condition. This processing difficulty for RC continuations is reduced but, crucially, not fully eliminated when the linguistic context provides two referents (e.g., *two women*) that can be distinguished by the RC (*the woman that X*). CC disambiguation also shows an interaction with context, especially in the sentence-final region, but it goes in the opposite direction compared to the RC condition: CC disambiguation leads to longer first-pass reading times and more first-pass regressions when the context provides two non-unique referents, presumably because the use of the definite NP in the ambiguous region (*He told the woman ...*) is not properly licensed in this configuration.

Figure 4 shows that the proportion of positive acceptability ratings is lower on average for trials in which participants regressed at least once. Regarding the effect of context, RC disambiguation is less acceptable when there is only one referent, while CC disambiguation shows no difference in acceptability between the different contexts.

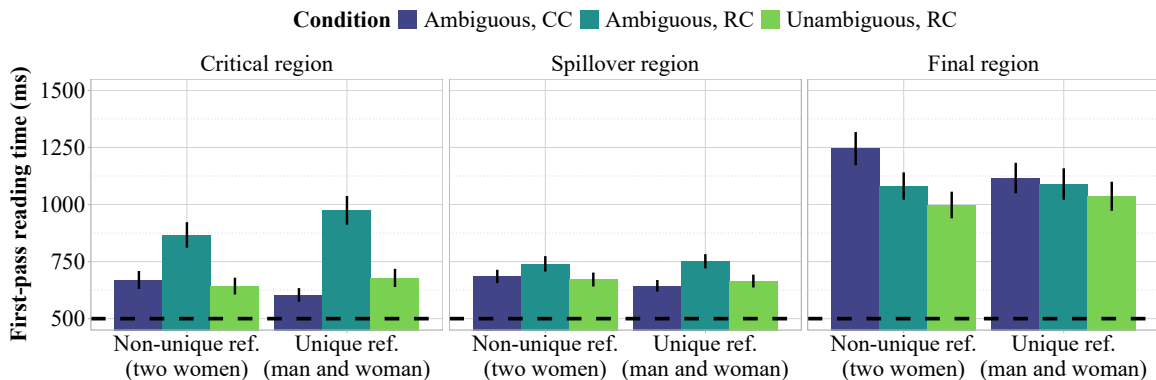


Figure 2: Length-corrected first-pass reading times by context, condition, and region. Note that the y-axis starts at 500 ms (dashed horizontal line). Error bars show 95% confidence intervals.

Overall, the early influence of context on measures of garden-pathing is more compatible with the context-sensitive attachment hypothesis (Altmann et al., 1992) than with the context-insensitive attachment hypothesis. Even though RC disambiguation leads to slower first-pass reading times and more regressions overall, providing non-unique as opposed to unique referents (two women versus a man and a woman) in the discourse context decreases reading times and regression proportions in the ambiguous RC condition, consistent with context-sensitive attachment. In addition, and also consistent with the context-sensitive account, when the context contains non-unique referents, CC disambiguation leads to longer reading times and more regressions from the sentence-final region, presumably because a uniqueness presupposition is violated when the intended referent of the woman is not identifiable (He told ?[the woman] [that X]). However, this violation in the CC condition does not appear to result in a lower proportion of end-of-trial acceptability judgments. By contrast, mismatching RC disambiguation does result in a lower acceptability judgement.

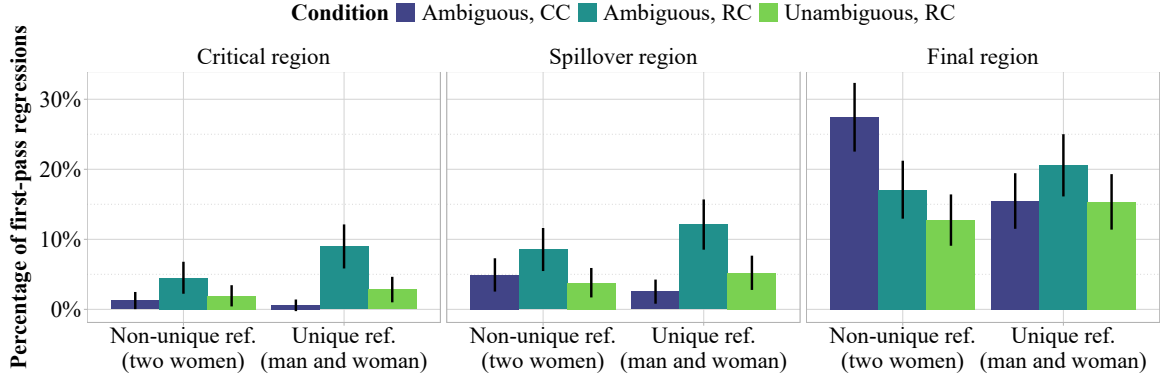


Figure 3: Percentage of first-pass regressions by context, condition, and region. Error bars show 95% confidence intervals.

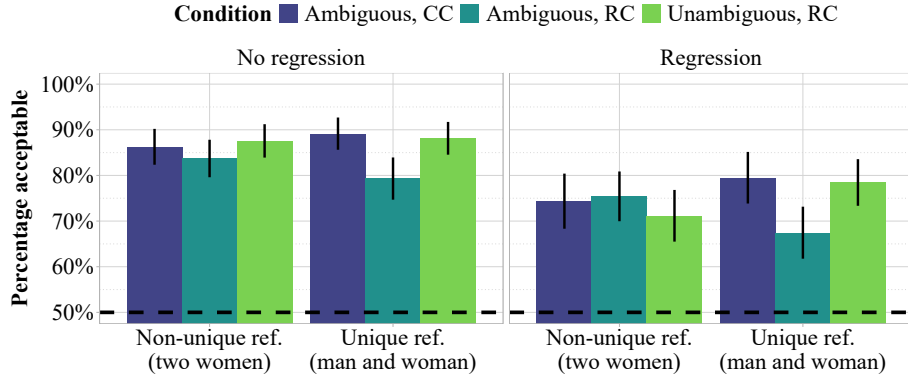


Figure 4: Percentage of positive acceptability judgments by context, condition, and presence or absence of at least one regression in the trial. Error bars show 95% confidence intervals.

Despite the presence of a context $\times$ disambiguation interaction, proponents of the context-insensitive attachment hypothesis could argue that, in matching contexts, first-pass reading times and regression probabilities for RC disambiguation should have been *lower* than for CC disambiguation if attachment was really context-sensitive. Instead, the opposite pattern is seen (see Figures 2 and 3). This suggests that first-pass attachment choices might not be influenced by context. But this is only an argument in favor of a general anti-RC bias, which we would agree probably exists (see also Staub et al., 2018). Crucially, under a nondeterministic view of parsing, the existence of such a bias does not preclude the possibility that the parser sometimes — or even often — adopts an RC analysis when it is supported by the context. This nondeterministic view arguably offers a more straightforward account of the data than assuming a deterministic parser that always initially adopts the CC analysis but can quickly use discourse information to reanalyze the structure, along the lines of Frazier’s classic garden-path model (Frazier, 1979).

In order to shed more light on the role of nondeterminism in the parsing process, we now fit an extended version the multinomial processing tree (MPT) model of Paape and Vasishth (2022b) to our data. As explained above, the model explicitly assumes that attachment choices are probabilistic, and predicts a mixture distribution of reanalyzed and non-reanalyzed trials in the garden-path condition. The model fit thus allows for a more fine-grained picture of the latent processes involved in garden-pathing compared to computing simple differences between the means of the experimental conditions. By exploring the effect of the context manipulation on the different stages of the processing cascade, we can draw more informative conclusions about which stages of the parsing mechanism are affected by contextual match.

In order to understand a model’s relative performance, it is important to pit it against competing models that may furnish potentially more parsimonious explanations for the observed patterns in the data. Accordingly, we also compare the MPT model to simpler, one-stage models. In this regard, because surprisal has been argued to account for a wide range of processing effects in psycholinguistics, including garden-pathing, (Hale, 2001; Levy, 2008), a crucial comparison is with models based solely on the disambiguating region’s surprisal, computed using large language models.

Computational modeling results

Parameter estimation

When fitting the MPT model, the probability and cost parameters are estimated based on the reading times at the disambiguating and spillover regions, the presence or absence of a first-pass regression, and the response to the end-of-trial acceptability judgment. The same factor coding as for the previous statistical analyses was used for the MPT model, which was implemented in Stan (Stan Development Team, 2024). An additional contrast was defined for the factor contextual MATCH, with context-disambiguation match coded as 1 and context-disambiguation mismatch coded as -1 .

The predictor structure across all parameters is as follows:

$$\begin{aligned} p_gp &\sim context + disambiguation + context : disambiguation \\ p_triage &\sim disambiguation + match \\ p_covert_success &\sim disambiguation + match \\ p_overt_success &\sim disambiguation + match^3 \\ covert_reanalysis_cost &\sim disambiguation + match \end{aligned}$$

The population-level probability p_bias was fixed at the posterior mean estimated by Paape and Vasishth (2022b) (79% “accept”-bias) but allowed to vary between participants. This choice was made because letting the parameter vary freely led to problems with parameter recovery in simulation studies. Four MCMC chains with 2000 iterations each were run using the CmdStanR interface (Gabry et al., 2024), 1000 of which were discarded as burn-in. $\hat{R}$ values were monitored for possible indications of non-convergence. We report 95% percentile-based credible intervals for each parameter.

Figure 5 plots the estimates of the intercept parameters for each process probability in the MPT; given our factor coding, the intercept corresponds to the mean process probability in the RC disambiguation condition across matching and mismatching contexts. Figure 6 plots the slope parameters for the process probabilities. Figure 7 plots the intercept and slope parameters for the process costs.

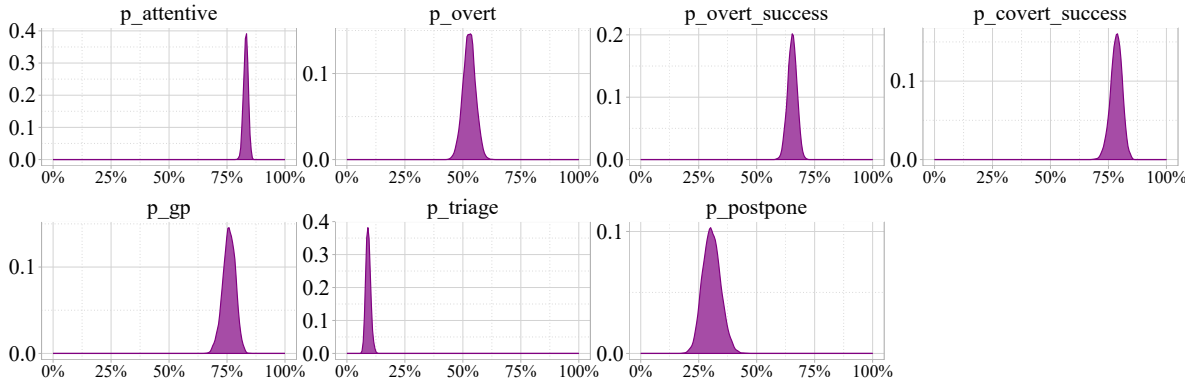


Figure 5: Posterior density plots of the MPT parameters (intercepts only).

The parameter estimates can be interpreted as follows:

1. Participants are reading attentively in about 80% of trials on average. Triage only occurs rarely (about 10% of the time), and about equally across all conditions.

³Rereading patterns from the scanpath analysis (see above) were also entered into the model as predictors of overt reanalysis success, but the results were inconclusive.

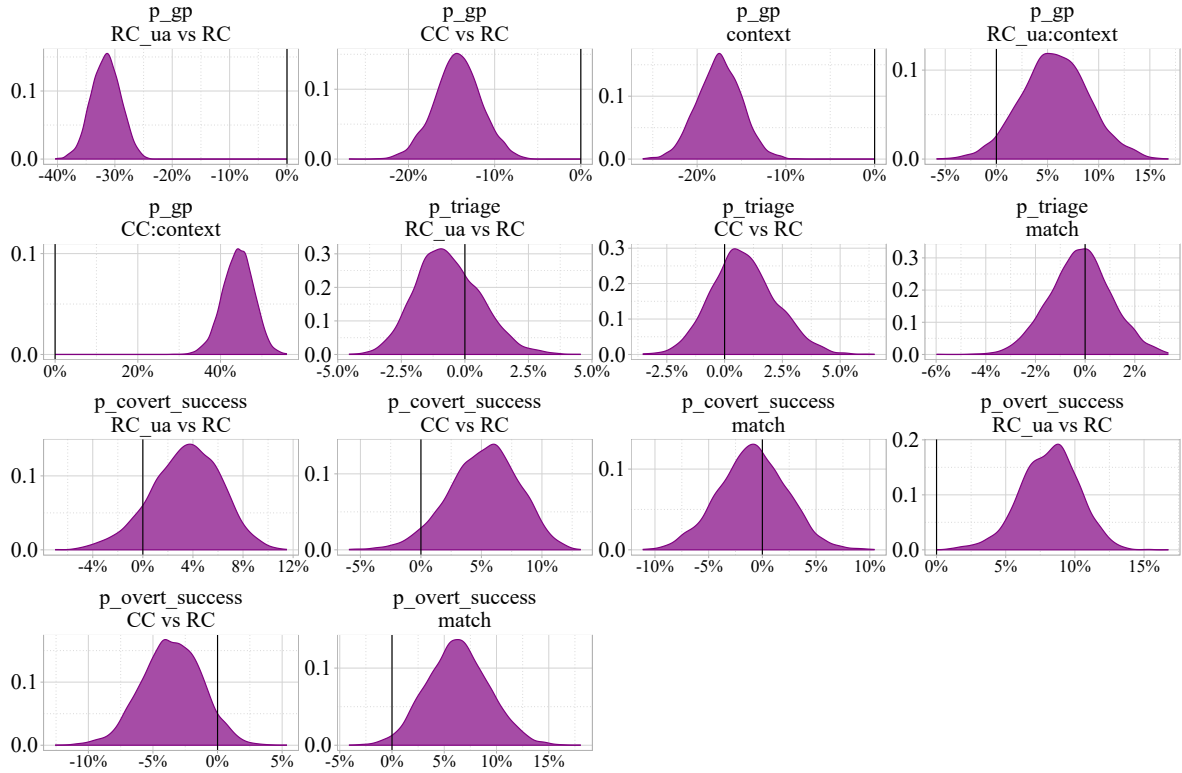


Figure 6: Posterior density plots of the MPT parameters (slopes only).

2. On average across contexts, garden-pathing occurs in about 75% of ambiguous RC trials but is reduced by about 30% in unambiguous RC trials, and by about 15% in ambiguous CC trials.
3. The probability of garden-pathing is strongly affected by context: Non-unique referents versus unique referents (*two women* vs. *man and woman*) decrease the probability of garden-pathing by about 18% in ambiguous RC trials but increase the probability of garden-pathing by about 45% in ambiguous CC trials.
4. Participants engage in overt reanalysis and covert reanalysis about equally often. Overt reanalysis succeeds in about two thirds of ambiguous RC trials while covert reanalysis succeeds in about 75% of ambiguous RC trials.
5. Pure garden-pathing with subsequent triage is rare but not very costly (about 25–45 ms), whereas covert, in-situ reanalysis is costly (about 330 ms).
6. Covert reanalysis is more costly (by about 170–230 ms) and more likely to fail for ambiguous RC trials compared to unambiguous RC trials and CC trials. The success probability of overt but not covert reanalysis is higher, and covert reanalysis is less costly, when the context matches the disambiguation.

We now turn to model comparisons, in order to establish whether the MPT model provides improved predictive fit to the data compared to mathematically simpler alternative models, including models that use surprisal as the sole predictor of garden-path cost.

Model comparisons using cross-validation

It is possible that the MPT model of garden-pathing and reanalysis does not perform as well as competing accounts that make fewer assumptions. If a simpler model can account for the data equally well, then that model should be preferred on grounds of parsimony. As stated in the introduction, one candidate class of such simple, wide-coverage models is based on surprisal, that is, the negative log probability of a word in context (Hale, 2001; Levy, 2008).

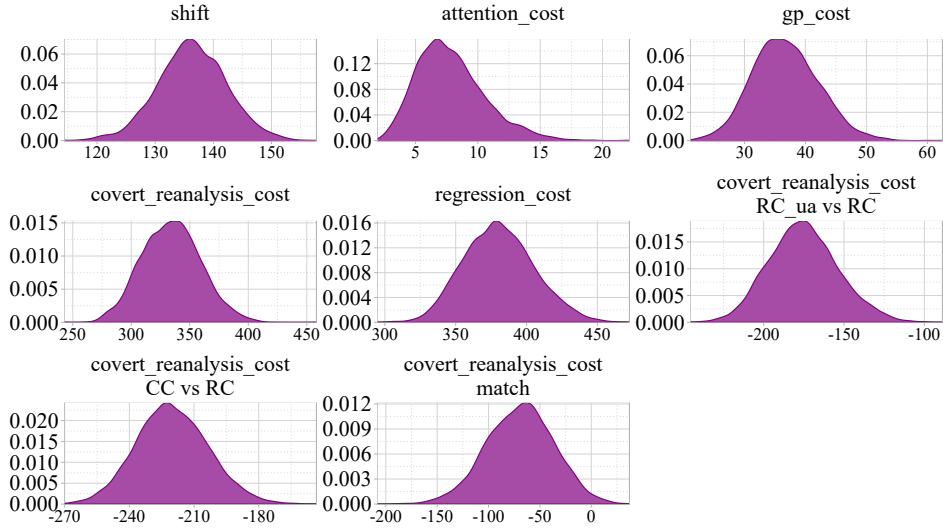


Figure 7: Posterior density plots of the MPT cost parameters.

Surprisal as computed from a large language model (LLM) has previously been found to underpredict the magnitude of human garden-path effects (Van Schijndel and Linzen, 2021; Huang et al., 2024). Speculatively, this may be due to the fact that LLMs lack the specialized and costly mechanisms for syntactic reanalysis that have classically been assumed in psycholinguistic theory (e.g., Frazier, 1979; Ferreira and Henderson, 1998). Surprisal-based models must attribute any cost associated with disambiguation towards a dispreferred syntactic alternative exclusively to changed predictions based on preceding words. Garden-path sentences have no special status under surprisal theory, and should conform to the assumed mapping between surprisal and processing difficulty just like any other sentence, contrary to the results of Van Schijndel and Linzen (2021) and Huang et al. (2024). Nevertheless, given the aforementioned success of LLM-derived surprisal as a predictor of human processing difficulty, as well as the in-principle possibility that an interaction between linguistic context and the preferred disambiguation of the CC/RC ambiguity can be learned from large-scale training data, it is important to evaluate the performance of LLM-derived surprisal in terms of predicting human data in the current setting, and to compare its predictive performance against that of the MPT model.

An important test of the different models considered here is their *predictive* performance: rather than asking how well each model approximates the data it was fitted to (“in-sample” fit), we evaluate how well it fits *unseen* data (“out-of-sample,” predictive fit). Predictive fit can be evaluated by means of cross-validation, a standard method in machine learning. In cross-validation, a subset of the data is withheld from the model during fitting. After the parameters are estimated, the joint density of the unseen data given the parameters is computed. This process is repeated multiple times with different subsets of the data being withheld, until the entire data set is covered. The model under which the unseen data are most likely has the best predictive performance. Given the computational cost of refitting complex models several times, the process can be approximated by methods such as PSIS-LOO (Vehtari et al., 2017), which makes use of importance sampling (Gelfand et al., 1992) to save computational resources. An important advantage of cross-validation is that the approach automatically penalizes overfitting, because overfitted models will have worse out-of-sample fit (Hawkins, 2004).

In order to evaluate LLM surprisal as a predictor of the context-disambiguation interaction, we chose a convenience sample of models available from HuggingFace (<https://huggingface.co>) with different numbers of parameters and context lengths: GPT2-1.5B (Radford et al., 2019), LLAMA2-7B (Touvron et al., 2023), MISTRAL-7B (Jiang et al., 2023), and FALCON-11B (Malartic et al., 2024). GPT2-1.5B has a context length of 1024 tokens, LLAMA2-7B has a context length of 4096 tokens, and MISTRAL-7B and FALCON-11B each have a context length of 8192 tokens.

As a first step, in order to see whether LLM surprisal can, in principle, account for the interaction between context and disambiguation in reading times, we computed the surprisal of each of the LLMs for the critical and spillover regions of each experimental sentence in each condition.⁴ The surprisal values were centered and scaled, and entered as the sole predictor of first-pass reading times at each region in a linear mixed-effects model using a lognormal likelihood. The predicted condition means — generated using the `predict()` function of the `lme4` package (Bates et al., 2015) — are shown in Figure 8.

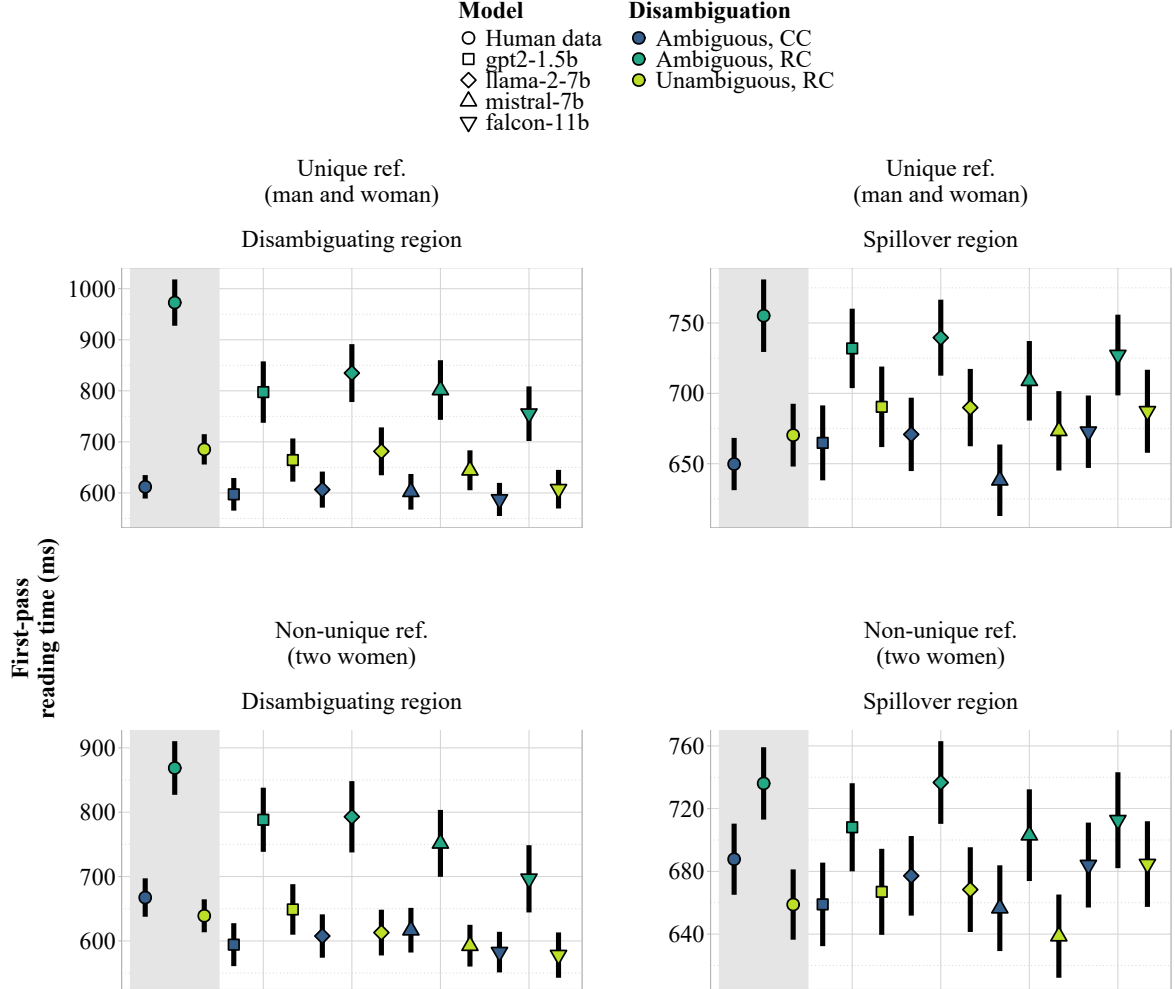


Figure 8: Human data (gray bar) versus model-predicted first-pass reading times at the disambiguating region and the spillover region for different LLMs. Error bars show 95% confidence intervals.

As Figure 8 shows, the surprisal models mostly capture the qualitative patterns in the data, but there are also discrepancies: the surprisal-based models underpredict the difficulty caused by CC disambiguations in contexts with non-unique referents, and underpredict the difficulty caused by RC disambiguations in contexts with unique referents. These results suggest that LLM surprisal is an adequate qualitative predictor of the context $\times$ disambiguation interaction in the current reading time data but not necessarily a good quantitative predictor.

For the cross-validation step, multivariate versions of the LLM surprisal models were fitted to predict reading times and trial type (see below). Cross-validation was carried out using the `loo` package in R (Vehtari et al., 2015). As a baseline “null” model, we fitted a multivariate model to reading times at the critical and spillover regions using a lognormal likelihood, and to the trial category (accept/no regression; accept/regression; reject/no regression; reject/regression) using a categorical likelihood. The baseline model has only an intercept and random adjustments by item and by subject, as well as a slope for (scaled) region length for reading times. The other reference model is the “factorial” model, which is identical to the baseline model but adds main effects and interactions for the factors

⁴Recall that each region contained several words, so that the summed surprisal of the entire string was computed.

CONTEXT and DISAMBIGUATION for all dependent variables. The surprisal-based models are similar to the multivariate categorical model, but use item- and condition-specific LLM surprisal values instead of the categorical predictors. The surprisal slope is hierarchically adjusted by participant. Neither the factorial model nor the LLM surprisal models are constrained with regard to effect directions, both for reading times and for trial type, and uninformative priors are used.

In addition to the “full” MPT model reported above and the four surprisal-based models, the following additional models were also fitted for comparison:

- MPT-NO-TRIAGE: The “full” MPT model without triage.
- MPT-SIMPLE: An MPT model including only probabilistic garden-pathing and non-optional reanalysis, but no inattention and no triage.
- FACTORIAL-INATTENTION: The factorial model with an additional mixture component intended to capture inattentive trials, analogous to the $p_attentive$ parameter of the MPT model.

Figure 9 shows the cross-validation results. The quantity of interest is the model’s expected log pointwise predictive density for the unseen data ($\widehat{elpd}$), summed over the entire data set, with higher values of $\widehat{elpd}$ indicating better predictive fit. This quantity has an associated standard error that can be used to calculate confidence intervals, and thus to evaluate whether two models differ significantly in their predictive fit (Vehtari et al., 2017).

As Figure 9 shows, all models significantly outperform the baseline “null” model in terms of predictive fit. The differences between the factorial model and the LLM surprisal models, as well as between the different LLM surprisal models, are comparatively small, though there is a slight trend towards surprisal from larger LLMs showing better predictive performance. By contrast, the “full” MPT model significantly outperforms the surprisal-based and factorial models by a factor of about 2.5. Removing the triage process from the MPT model results in a massive increase in predictive fit. On the other hand, removing both inattention *and* triage from the MPT, so that only garden-pathing and reanalysis remain, significantly reduces predictive fit. Meanwhile, adding a mixture component to the factorial model to capture inattentive trials increases its predictive fit considerably, even slightly above that of most of the simplest MPT model.

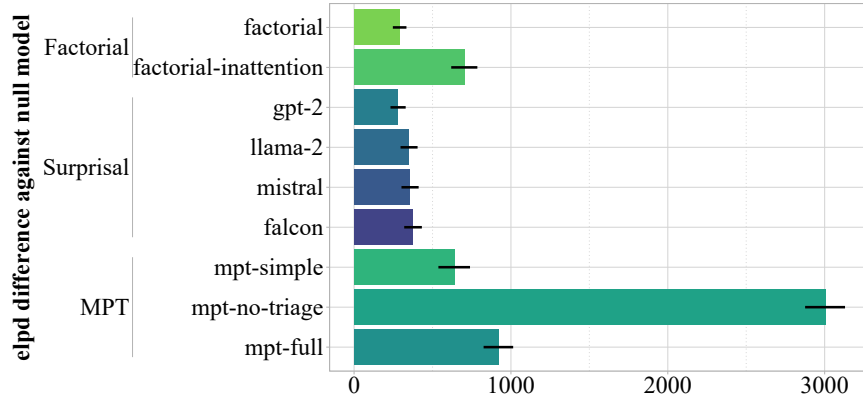


Figure 9: Model comparison results based on PSIS-LOO. Higher values of $\widehat{elpd}$ indicate better predictive fit. Error bars show 95% confidence intervals.

Where is the predictive advantage of the no-triage MPT model coming from? One way of exploring this question is to plot the individual observations in the data set together with their $\widehat{elpd}$ values under each model (Nicenboim and Vasisht, 2018). We take a slightly different approach: in Figure 10, data points are ordered by trial type — accept/reject and no regression/regression — and by first-pass reading times at the critical and spillover regions, and each data point is colored according to the model that best predicts it. For reasons of visibility, we have limited the visual comparison to the factorial, full-MPT, no-triage MPT and GPT2-surprisal models. As Figure 10 shows, the main advantage of the no-triage MPT model clearly comes from its superior predictive performance in trials in which the sentence is rejected but where regressions are absent.

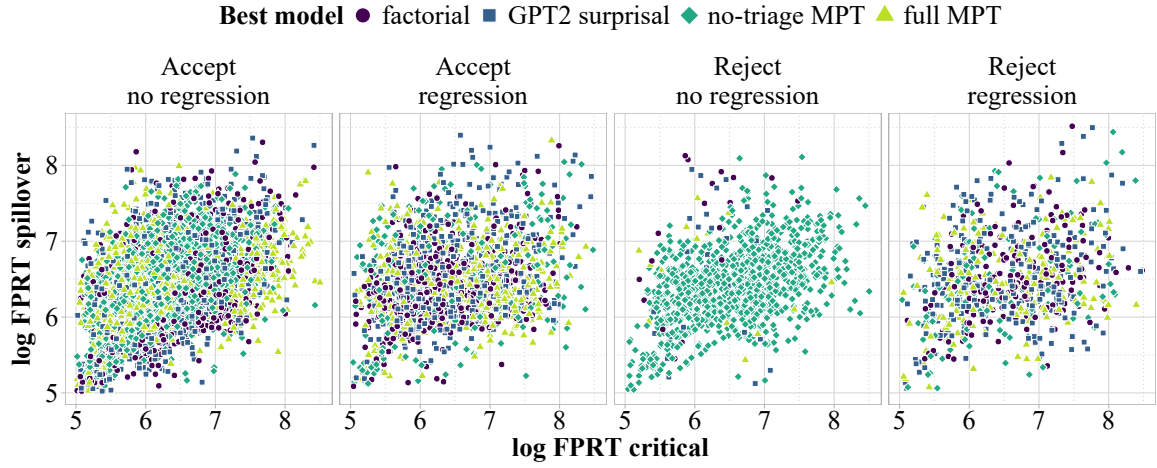


Figure 10: Best-predictive model by trial type and log first-pass reading time at the critical and spillover regions. Point colors indicate the model with the highest $\widehat{elpd}$ for the observation.

Discussion

The parameter estimates of the full MPT model are in line with the descriptive results, with earlier published results, and with psycholinguistic theory: the ambiguous conditions result in a higher probability of garden-pathing than the unambiguous conditions, garden-pathing is influenced by context, and disambiguation towards the RC structure is more difficult than disambiguation towards the CC structure. Importantly, due to the mixture assumption inherent in the MPT model, the parameter estimates yield more information than the simple comparison of condition means: by assuming a cascade of latent cognitive processes, the effect of context on the initial garden-path probability can be distinguished from the effect on reanalysis costs and success probabilities, which combine nondeterministically to yield the observed reading times, regression probabilities, and acceptability judgments.

The cross-validation results show that the increased complexity of the MPT model in terms of the number of fitted parameters compared to a standard factorial model goes along with a significant improvement in predictive fit to out-of-sample data. Furthermore, an MPT model that removes all latent processes and their associated mixture components except costly reanalysis has about the same predictive fit as standard factorial model extended solely with an inattention component — that is, a second mixture component with a faster mean reading time, lower variance, and no effects of the experimental manipulations. This suggests that the inattention component plays a crucial role in determining the predictive fit of the MPT models. On the other hand, the more complex MPT models significantly outperform the factorial-inattention model, suggesting that the additional assumption of distinct garden-pathing and reanalysis components increases predictive fit above and beyond the inattention component alone.

The finding that the no-triage MPT model outperforms the full MPT model for trials without regressions is somewhat surprising given that triage is supposed to account precisely for trials in which readers are garden-pathed but do not reanalyze and instead reject the sentence. However, there are two other routes through the MPT that generate reject/no regression trials: the *inattention* $\rightarrow$ *biased guess* route and the *garden-pathing* $\rightarrow$ *covert reanalysis* route. The parameter estimates indicate that the no-triage model estimates a lower cost for covert reanalysis than the full MPT model, presumably to account for relatively “fast” rejections in the absence of regressions. Furthermore, one aim of the cross-validation technique as applied to mixture models such as the different MPT versions is to evaluate how many mixture components are actually needed to accurately predict unseen data (e.g., Smyth, 2000; McLachlan and Rathnayake, 2014). In our case, the answer is that the triage component is not needed. This conclusion is at odds with the findings of Paape and Vasishth (2022b), which indicated that the MPT model with triage outperformed the one without triage. This mismatch may speculatively be due to our extension of the model with a spillover component: when only the critical region is analyzed, trials in which reanalysis occurs (but fails) at the spillover region may be

incorrectly grouped as triaged trials. Alternatively, the changed experimental setup in our study relative to Paape and Vasishth (2022b), which required participants to judge sentences in context rather than out-of-the-blue, may have contributed to fewer triaged trials.

In summary, our results suggest that the assumption of a tripartite mixture of inattentive trials, non-garden-path trials in which the correct analysis is chosen right away, and reanalysis trials lets these models predict the human data more accurately than single-component models, including surprisal-based models.

General discussion

We have presented experimental results from bidirectional self-paced reading that support context-sensitive attachment preferences in English complement clause/relative clause ambiguities (e.g., *He asked the woman that ...*). For sentences that were disambiguated towards an RC analysis, first-pass reading times at the point of disambiguation were shorter, and there were fewer first-pass regressions, when the extrasentential linguistic context provided non-unique referents that could be distinguished by an RC (e.g., *two women* as opposed to *a man and a woman*). Furthermore, the data showed that CC continuations also come with a cost in non-supporting contexts, in that they cause longer reading times and more regressions from the sentence-final region. This latter pattern has, to our knowledge, not been reported before. Importantly, the results suggest a theoretical compromise, in that first-pass parsing is not *fully* context-sensitive, but definitely not entirely context-insensitive either: RC disambiguation always led to more processing difficulty on average at the disambiguating region than CC disambiguation, but there was also a clear effect of context on the earliest reading measures.

Despite their obvious informativity with regard to incremental processing, reading data are only one of several sources of information about the processes involved in garden-path processing. For instance, acceptability judgments provide useful information with regard to whether a grammatical structure was ultimately derived (Warner and Glass, 1987).

To shed more light on where exactly in the cascade of cognitive processes context exerts its effect on parsing, we estimated parameters for an extended version of the multinomial processing tree (MPT) model of garden-pathing proposed by Paape and Vasishth (2022b), which aims to jointly account for reading measures and acceptability judgments. The model separates parsing into attention, initial structural choice, and optional reanalysis. We extended the model by adding a spillover component that allows covert reanalysis to be performed at the spillover region as opposed to the critical region, and also included regressions from any post-critical region to capture attempts at overt reanalysis. Our parameter estimates suggest that the initial probability of garden-pathing as well as the cost and success probabilities of reanalysis are all affected by context to some extent: a matching context reduces the probability of garden-pathing, reduces the cost of “covert,” in-situ revision, and increases the probability of successful reanalysis through rereading. Again, the results support a synthesis of existing accounts: context affects initial parsing decisions but also aids reanalysis.

An additional question we sought to answer was whether the context-disambiguation interaction could plausibly be explained by differences in surprisal, that is, by the assumption that a CC or RC continuation may be more expected in a supporting context than in a non-supporting one. The surprisal values computed from four different large language models mostly accounted for the qualitative patterns in human reading times but did not fully capture the quantitative patterns. These mismatches were evident despite the fact that the mixed-effects models with the surprisal predictor were only fitted to the garden-path sentences. In other words, surprisal was given a very generous opportunity to predict the reading times. This is in contrast to earlier studies (Van Schijndel and Linzen, 2021; Huang et al., 2024) set a much higher bar for surprisal by first using filler sentences to obtain a slope for surprisal, and then multiplying this slope with the surprisal values of the garden-path sentences.

Comparing the predictive fit of different variants of the MPT model of Paape and Vasishth (2022b) against the surprisal-based models, against a standard factorial model, and against a factorial model with an inattention component showed that the “full” and no-triage MPT models consistently outperformed the other models. Surprisingly, the no-triage MPT model, that is, the MPT model without the assumption that garden-path sentences are sometimes outright rejected without reanalysis (Meng and Bader, 2000; Fodor and Inoue, 2000), performed best. For the factorial model, adding an inattention component measurably improved predictive fit, showing that the assumption of

occasional inattention or “mindless reading” (e.g., Smallwood, 2011; Schad et al., 2012) may yield a more complete picture participants’ actual mental state in reading experiments, especially when the perceived stakes are relatively low (Christianson et al., 2022).

Why do the MPT models outperform the surprisal-based models in terms of predictive fit to human data? In earlier work that discussed the failure of surprisal to account for the magnitude of human garden-path effects, researchers have suggested that the mismatch may be due to a costly reanalysis process in a serial, two-step parsing architecture that surprisal fails to account for (Van Schijndel and Linzen, 2021; Huang et al., 2024; Arehalli et al., 2023; Huang and Dillon, 2023). By contrast, the mixture assumption inherent to the MPT architecture straightforwardly predicts that the distribution of reading times at the disambiguating region will underlyingly be bimodal, with reanalyzed trials being distinguishable from non-reanalyzed trials.

Finally, nothing precludes the formulation of a multi-component mixture model that still assumes a role for surprisal. As our model comparison results have shown, item-level surprisal can substitute for categorical factor codings based on the experimental conditions, which surprisal, of course, correlates with. This means that item-level surprisal can simply be added as a predictor to any of the MPT parameters instead of the factorial predictors, so that it can affect the initial garden-path probability as well as the cost and/or success of reanalysis. Due to the intrinsic connection between reading measures and acceptability judgments assumed by the MPT, such a model would also explicitly connect participants’ judgments with surprisal, a relationship that has historically been difficult to establish and to theoretically account for (e.g., Lu et al., 2024). We have refrained from implementing such a model here, because we’re not sure how to interpret it within the broader surprisal framework. Surprisal models are usually taken to be single-stage, non-serial models (Van Schijndel and Linzen, 2021), so that it becomes difficult to envision a role for serial, multi-stage notions such as reanalysis and triage. Nevertheless, under appropriately justified assumptions, such a model may provide the most empirically adequate fit to human data.

Conclusion

We present a multinomial processing tree (MPT) model fit to data from a large-sample bidirectional self-paced reading study. The data and modeling show that (i) context influences the probability of choosing a matching syntactic attachment during first-pass processing; however, (ii) the influence of context does not completely override structural biases; and (iii) contextual match also influences the cost and success rate of syntactic reanalysis. Model comparison using cross-validation shows that the MPT model outperforms models using surprisal computed from large language models.

Acknowledgments

This work was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – Project number 428960187 – PI: Dario Paape.

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1 Appendix: Statistical results

All data analysis was carried out in R (R Core Team, 2022), and models were fitted with the *brms* package (Bürkner, 2017, 2018). Reading measures were computed using the *em2* package (Logačev and Vasishth, 2013). First-pass reading times below 150 ms and above 5000 ms were discarded, resulting in a loss of 5% of the data. We analyzed first-pass reading times and the percentage of first-pass regressions as the earliest possible measures potentially affected by garden-pathing that can be computed for BSPR. A sum contrast was defined for the two-level factor *CONTEXT* such that contexts with unique referents (e.g., *a man and a woman*) were coded as -1 and contexts with non-unique referents (e.g., *two women*) were coded as 1 . A treatment contrast was defined for the three-level factor *DISAMBIGUATION*, so that the ambiguous CC condition and the unambiguous RC condition are compared to the ambiguous RC condition as the baseline. Region length in characters was added as a centered predictor, as the average length of the disambiguating and spillover regions was different between the experimental conditions. A lognormal likelihood was assumed for FPRTs while a Bernoulli likelihood was assumed for first-pass regressions. We report 95% percentile-based credible intervals and Bayes factors in favor of the alternative hypothesis (BF10) for each comparison, using mildly informative priors for the effects of interest with moderate effect sizes (± 80 ms, $\pm 15\%$).⁵

Table 2 shows the results of the statistical analyses for the reading measures. Table 3 shows the results of the statistical analysis for the acceptability judgments.

⁵Bayes factors were computed with the *bayestestR* package (Makowski et al., 2019). For guidance regarding the interpretation of Bayes factors, see e.g., Lee and Wagenmakers (2014). We treat Bayes factors above 3 in favor of the alternative hypothesis as evidence of an effect. Because Bayes factors can be influenced by the prior specification (Kass and Raftery, 1995), we use mildly informative priors, especially for the target parameter being investigated (Schad et al., 2022).

Critical region						
	FPRT (ms)			% regressions		
	CI low	CI high	BF10	CI low	CI high	BF10
ctxt	-84	-39	> 1000	-4%	-1%	46.68
cc	-196	-65	467.40	-6%	1%	2.16
rc_ua	-200	-73	611.36	-6%	1%	2.07
ctxt:cc	78	139	> 1000	0%	6%	10.57
ctxt:rc_ua	7	64	4.73	-1%	3%	0.86
Spillover region						
	FPRT (ms)			% regressions		
	CI low	CI high	BF10	CI low	CI high	BF10
ctxt	-31	8	0.28	-3%	0%	4.44
cc	-133	-17	11.36	-10%	-1%	17.58
rc_ua	-143	-24	19.35	-7%	1%	3.33
ctxt:cc	22	70	141.58	2%	7%	107.95
ctxt:rc_ua	-17	30	0.20	-2%	3%	0.56
Final region						
	FPRT (ms)			% regressions		
	CI low	CI high	BF10	CI low	CI high	BF10
ctxt	-18	44	0.25	-4%	0%	2.02
cc	14	177	6.92	-4%	5%	0.53
rc_ua	-184	-20	9.39	-14%	-4%	170.61
ctxt:cc	25	128	19.62	8%	15%	> 1000
ctxt:rc_ua	-60	28	0.31	-4%	3%	0.38

Table 2: 95% credible intervals and Bayes factors for first-pass reading times and percentage of first-pass regressions by region. ctxt: Context, cc: CC vs RC contrast, rc_ua: unambiguous vs ambiguous RC contrast. Bayes factors >3 shown in bold. The table gives back-transformed estimates obtained by back-transforming the edges of the 95% credible intervals on the log/logit scale.

	CI low	CI high	BF10
ctxt	3%	9%	324.85
cc	-4%	12%	1.12
rc_ua	4%	22%	43.16
reg	-15%	-8%	> 1000
ctxt:cc	-19%	-9%	> 1000
ctxt:rc_ua	-6%	4%	0.43
ctxt:reg	-3%	3%	0.25
cc:reg	-6%	4%	0.46
rc_ua:reg	-9%	1%	1.58
ctxt:cc:reg	-5%	4%	0.42
ctxt:rc_ua:reg	-9%	0%	2.00

Table 3: 95% credible intervals and Bayes factors for proportions of positive acceptability judgments. ctxt: Context, cc: CC vs RC contrast, rc_ua: unambiguous vs ambiguous RC contrast. Bayes factors >3 shown in bold. The table gives back-transformed estimates obtained by back-transforming the edges of the 95% credible intervals on the logit scale.